THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 8 — MATHS FULL SYLLABUS PAPER SET 03

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. How many non-square numbers lie between 7^2 and 8^2? 1

(a) 7 **(b)** 14 **(c)** 15 **(d)** 13

2. The smallest number by which 72 must be multiplied to get a perfect square is: 1

(a) 2 **(b)** 4 **(c)** 3 **(d)** 6

3. $5^2 \times 5^7$ equals: 1

(a) 25^9 **(b)** 5^{-5} **(c)** 5^{14} **(d)** 5^9

4. The decimal number 26 written in base 3 is: 1

(a) 1000 **(b)** 222 **(c)** 1001 **(d)** 221

5. A quadrilateral in which both pairs of opposite angles are equal is a: 1

(a) trapezium **(b)** parallelogram **(c)** kite **(d)** scalene quad

6. One angle of a parallelogram is 65°. Its adjacent angle is: 1

(a) 115° **(b)** 25° **(c)** 230° **(d)** 65°

7. Is 8109 divisible by 9? 1

(a) Yes **(b)** Only if the last digit is even

(c) No **(d)** Only if the number is prime

8. The coefficient of x in $(x+5)(x+2)$ is: 1

(a) 7 **(b)** 10 **(c)** 3 **(d)** 5

9. If 4 : 9 = 20 : x, then x equals: 1

(a) 46 **(b)** 48 **(c)** 44 **(d)** 45

10. ₹ 840 is divided between A and B in the ratio 3 : 4. A's share is: 1

(a) ₹ 480 **(b)** ₹ 600 **(c)** ₹ 420 **(d)** ₹ 360

11. ₹ 1000 at 10% compound interest annually for 2 years gives an amount of: 1

(a) ₹ 1100 **(b)** ₹ 1200 **(c)** ₹ 1220 **(d)** ₹ 1210

12. An article bought for ₹ 250 is sold for ₹ 200. The loss percent is: 1

(a) 25% **(b)** 20% **(c)** 50% **(d)** 10%

13. The side of a square whose area is double that of a square of side a is: 1

(a) $a\sqrt{2}$ **(b)** $2a$ **(c)** $a+2$ **(d)** $a\sqrt{3}$

14. The diagonal of a rectangle 8 cm by 15 cm is: 1

(a) 17 cm **(b)** 23 cm **(c)** 13 cm **(d)** 19 cm

15. A map scale 1 : 100000 means 1 cm on the map is how many km on the ground? 1

(a) 10 km **(b)** 0.1 km **(c)** 100 km **(d)** 1 km

16. A solid has 5 faces and 6 vertices. Using $F + V - E = 2$, the number of edges is: 1

(a) 9 **(b)** 7 **(c)** 13 **(d)** 11

17. The data 10, 10, 10, 10 has a mean of: 1

(a) 10 **(b)** 1 **(c)** 40 **(d)** 0

18. The two-digit number with tens digit 3 and ones digit 7 can be written as: 1

(a) $10 \times 7 + 3$ **(b)** $10 \times 3 + 7$ **(c)** 3×7 **(d)** $3 + 7$

19. The area of a trapezium with parallel sides 8 cm and 12 cm and height 5 cm is: 1

(a) 58 cm² **(b)** 96 cm² **(c)** 100 cm² **(d)** 50 cm²

20. A median of a triangle divides it into two parts of: 1

(a) area 1 : 2 **(b)** area 1 : 3 **(c)** equal area **(d)** unequal area

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 10–15 words each.

21. Find x if $5^x = 625$. 2

22. Why did Europeans call Hindu numerals 'Arabic numerals'? 2

23. What is the sum of the first 10 odd numbers? Which pattern does it follow? 2

24. Find the ratio of the perimeter of a square of side 6 cm to the side. 2

25. What is the side of a square whose diagonal is 14 cm? 2

26. 20 students eat in 12 days with a stock of food. How long does it last for 30 students? 2

27. A line graph shows temperature rising from 20°C to 30°C between 6 a.m. and 12 noon. Find the average rise per hour. 2

28. The diagonal of a rectangle splits it into two triangles. If the rectangle has area 84 cm², find the area of each triangle. 2

SECTION — C (Short Answer Type) 8 × 3 = 24

Answer in about 30–40 words each.

29. Compare the growth of squares and cubes: which is larger for 12, 12^2 or 12^3? By how much? 3

30. A rectangular park is 40 m by 30 m. Find the length of the diagonal path across it. 3

31. Find the area of a square of side 4.5 m using $(4 + 0.5)^2$. 3

32. Find the compound interest on ₹ 12000 at 6% per annum for 2 years. 3

33. A cube has edge 3 cm. Find its surface area and the number of such cubes needed to make a cuboid of dimensions $6 \times 3 \times 3$. 3

34. Find x in the grid trick: sum of 2×2 block starting at x equals 92. 3

35. A rectangular park measures 20 m by 21 m. A boy walks along the diagonal instead of two sides. How much distance does he save? 3

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OR

A man walks 12 m north and then 5 m east. How far is he from the start? 3

36. A trapezium has area 150 cm², parallel sides 12 cm and 18 cm. Find its height. 3

OR

A parallelogram has sides 13 cm and 20 cm and the height on 20 cm is 12 cm. Find its area and the height on the 13 cm side. 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 80–100 words each.

37. (i) Simplify $\frac{16^2 \times 2^3}{4^3}$. 5

(ii) Express the result as a power of 2.

OR

(i) Show that $(ab)^n = a^n b^n$ for $a = 2,\ b = 3,\ n = 3$. 5

(ii) Hence find $12^3 \div 6^3$.

38. (i) A phone bought for ₹ 15000 loses 20% value in the first year and 10% in the second. Find its value after 2 years. 5

(ii) Total depreciation percent.

OR

(i) Show that two successive increases of 10% equal one increase of 21%. 5

(ii) What about 10% increase followed by 10% decrease?

39. (i) If each of 9 observations is multiplied by 2, how do the mean and median change? 5

(ii) Illustrate with 3, 5, 7.

OR

(i) Find the median of 17 numbers where the sorted 9th value is 42. 5

(ii) Why?

40. **Case study:** A school ground is a trapezium with parallel sides 120 m and 80 m, and the non-parallel sides 50 m each. 5

(i) Find the height.

(ii) Find the area.

(iii) Find the cost of turf at ₹ 15 per m².

OR

(i) A square and a rectangle have equal perimeter 48 cm; the rectangle's length is 16 cm. Compare their areas. 5

(ii) What do you conclude?

END OF QUESTION PAPER — Total: 80 Marks